



Analog CMOS Circuit Design SS26

Practical Exercise IV – RC-Compensation

Group

Name Zhanyu Dong Sumit Kumar Patel

Matriculation Number 6159975 5992625

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Feedback

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Points **Mark**

Date **Signature**

No review without this standard form!



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1 T1

1.1 a

For the first stage, the output node N_1 is connected to the drain of the input transistor T_2 and the drain of the active-load transistor T_4 .

$$R_1 = r_{DS2} \parallel r_{DS4}$$

For the second stage, the output node N_2 is connected to the drain of the common-source transistor T_6 and the drain of the current-source load transistor T_7 .

$$R_2 = r_{DS6} \parallel r_{DS7}$$

At high frequency after Miller Pole Splitting.

$$R_{eff} = \frac{1}{gm_6}$$

1.2 b

The capacitance C_1 is the total capacitance from node N_1 to AC ground. At node N_1 , the important contributions are the drain-bulk and gate-drain capacitances of the first-stage output transistors, together with the input capacitance of transistor T_6 . Thus,

$$C_1 = C_{db2} + C_{db4} + C_{gd2} + C_{gd4} + C_{gs6}$$

$$C_2 = (C_L) + C_{db6} + C_{db7} + C_{gd7}$$

Miller coupling capacitance

$$Z_M = \frac{1}{sC_{gd6}}.$$

2 T2

2.1 a

$$I_{in} = \frac{v_{in}}{R_i}$$

in parallel with the resistor

$$R_i.$$

The current source I_{in} injects current into node 1. The node voltage at node 1 is denoted by V_1 . Therefore, v_{in} and V_1 must be distinguished.

2.2 b

At node 1

$$I_{in} = Y_1 V_1,$$

where

$$Y_1 = \frac{1}{R_i}.$$

At node 2, the node voltage is V_2 . This node is connected to R_1 , C_1 , the coupling impedance Z_M , and the first transconductance source $g_{m1}V_1$.

Define

$$Y_2 = \frac{1}{R_1} + sC_1$$

and

$$Y_M = \frac{1}{Z_M}.$$

The KCL at node 2 is

$$Y_2 V_2 + Y_M (V_2 - V_3) + g_{m1} V_1 = 0.$$

At node 3, the node voltage is $V_3 = v_{out}$. This node is connected to R_2 , C_2 , C_L , the coupling impedance Z_M , and the second transconductance source $g_{m6}V_2$.

Define

$$Y_3 = \frac{1}{R_2} + s(C_2 + C_L).$$

The KCL at node 3 is

$$Y_3 V_3 + Y_M(V_3 - V_2) + g_{m6} V_2 = 0.$$

2.3 c

The system of equations can be written as

$$\begin{bmatrix} \frac{v_{in}}{R_i} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} Y_1 & 0 & 0 \\ g_{m1} & Y_2 + Y_M & -Y_M \\ 0 & g_{m6} - Y_M & Y_3 + Y_M \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}.$$

with

$$Y_1 = \frac{1}{R_i}$$

$$Y_2 = \frac{1}{R_1} + sC_1$$

$$Y_3 = \frac{1}{R_2} + s(C_2 + C_L)$$

$$Y_M = \frac{1}{Z_M}$$

2.4 d

Matlab results:

```

1 V1/vin =
2 1
3
4 V2/vin =
5 -(R1 gm1 (C2 R2 s + CL R2 s + CM R2 s + 1))/(C1 R1 s +
   C2 R2 s + CL R2 s + CM R1 s + CM R2 s
6
7
2
2
2
2

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8      + CM R1 R2 gm6 s + C1 C2 R1 R2 s + C1 CL R1 R2 s +
      C1 CM R1 R2 s + C2 CM R1 R2 s + CL CM R1 R2 s
9
10     + 1)
11
12 V3/vin =
13 (R1 R2 gm1 (gm6 - CM s))/(C1 R1 s + C2 R2 s + CL R2 s +
      CM R1 s + CM R2 s + CM R1 R2 gm6 s
14
15           2                2                2
16     + C1 C2 R1 R2 s + C1 CL R1 R2 s + C1 CM R1 R2 s +
      C2 CM R1 R2 s + CL CM R1 R2 s + 1)
17
18 Numerator =
19 (-CM R1 R2 gm1) s + R1 R2 gm1 gm6
20
21 Denominator =
22 (C1 C2 R1 R2 + C1 CL R1 R2 + C1 CM R1 R2 + C2 CM R1 R2 +
      CL CM R1 R2)
23
24     2
25     s + (C1 R1 + C2 R2 + CL R2 + CM R1 + CM R2 + CM R1
      R2 gm6) s + 1
26
27
28 Check V3/vin - H_lecture =
29 0

```

$$\frac{V_1}{v_{in}} = 1$$

$$\frac{V_2}{v_{in}} = -\frac{R_1 g_{m1} [1 + s R_2 (C_2 + C_L + C_M)]}{p(s)}$$

$$\frac{v_{out}}{v_{in}} = \frac{R_1 R_2 g_{m1} g_{m6} \left(1 - \frac{s C_M}{g_{m6}}\right)}{p(s)}$$

$$p(s) = 1 + s [R_1 (C_1 + C_M) + R_2 (C_2 + C_L + C_M) + C_M g_{m6} R_1 R_2]$$

$$+ s^2 R_1 R_2 [C_1 (C_2 + C_L) + C_M (C_2 + C_L) + C_1 C_M]$$

3 T3: Determination of Zeros and Poles

3.1 a

The generic admittance Y_M is replaced by the admittance of a series $R_C - C_M$ compensation network.

$$Y_M = \frac{1}{Z_M} = \frac{1}{R_C + \frac{1}{sC_M}} = \frac{sC_M}{1 + sR_C C_M}.$$

We simulate in matlab and get

```

1      Numerator =
2      (CM R1 R2 Rc gm1 gm6 - CM R1 R2 gm1) s + R1 R2 gm1 gm6
3
4      Denominator =
5
6      (C1 C2 CM R1 R2 Rc + C1 CL CM R1 R2 Rc) s3 + (C1 C2 R1
7      R2 + C1 CL R1 R2 + C1 CM R1 R2 + C2 CM R1 R2
8      + CL CM R1 R2 + C1 CM R1 Rc + C2 CM R2 Rc + CL CM R2
9      Rc)
10
11     s2 + (C1 R1 + C2 R2 + CL R2 + CM R1 + CM R2 + CM Rc +
      CM R1 R2 gm6) s + 1

```

the transfer function can be written as

$$\frac{V_3}{V_1} = \frac{b_1 s + b_0}{a_3 s^3 + a_2 s^2 + a_1 s + a_0}.$$

$$b_0 = g_{m1} g_{m6} R_1 R_2$$

$$b_1 = g_{m1} R_1 R_2 C_M (g_{m6} R_C - 1).$$

$$a_0 = 1,$$

$$a_1 = R_1 C_1 + R_2 (C_2 + C_L) + C_M R_1 + C_M R_2 + C_M R_C + C_M R_1 R_2 g_{m6}.$$

$$a_2 = R_1 R_2 (C_1 C_2 + C_1 C_L + C_1 C_M + C_2 C_M + C_L C_M) \\ + C_1 C_M R_1 R_C + C_2 C_M R_2 R_C + C_L C_M R_2 R_C.$$

$$a_3 = R_C R_1 R_2 C_M C_1 (C_2 + C_L)$$

3.2 b

The numerator is a first-order polynomial,

$$b_1s + b_0.$$

Therefore, the transfer function exhibits one zero.

The denominator is a third-order polynomial,

$$a_3s^3 + a_2s^2 + a_1s + a_0.$$

Therefore, the transfer function generally exhibits three poles.

For purely Miller compensation,

$$R_C = 0.$$

In this case,

$$a_3 = R_C R_1 R_2 C_M C_1 (C_2 + C_L) = 0.$$

Therefore, for purely Miller compensation, the transfer function has two finite poles.

The zero for $R_C = 0$ becomes a right-half-plane zero

3.3 c

$$s_z = \frac{g_{m6}}{C_M(1 - g_{m6}R_C)}$$

For purely Miller compensation,

$$R_C = 0,$$

$$s_z = \frac{g_{m6}}{C_M}.$$

This zero is located in the right half-plane.

If

$$R_C = \frac{1}{g_{m6}},$$

then the denominator of the zero expression becomes zero and the zero is pushed to infinity.

If

$$R_C > \frac{1}{g_{m6}},$$

then

$$s_z < 0,$$

and the zero is moved to the left half-plane.

3.4 d

$$\left(1 - \frac{s}{p_1}\right) \left(1 - \frac{s}{p_2}\right) \left(1 - \frac{s}{p_3}\right) \approx 1 - \frac{1}{p_1}s + \frac{1}{p_1p_2}s^2 - \frac{1}{p_1p_2p_3}s^3.$$

Comparing coefficients with

$$1 + \frac{a_1}{a_0}s + \frac{a_2}{a_0}s^2 + \frac{a_3}{a_0}s^3,$$

we obtain

$$-\frac{1}{p_1} = \frac{a_1}{a_0},$$

$$\frac{1}{p_1p_2} = \frac{a_2}{a_0},$$

and

$$-\frac{1}{p_1p_2p_3} = \frac{a_3}{a_0}.$$

Therefore, the poles can be approximated by

$$p_1 \approx -\frac{a_0}{a_1}$$

$$p_2 \approx -\frac{a_1}{a_2}$$

$$p_3 \approx -\frac{a_2}{a_3}$$

3.5 e

In a_1 , the dominant term is

$$C_M g_{m6} R_1 R_2.$$

Therefore,

$$a_1 \approx C_M g_{m6} R_1 R_2.$$

Since

$$a_0 = 1,$$

the dominant pole is

$$p_1 \approx -\frac{a_0}{a_1}.$$

Thus,

$$p_1 \approx -\frac{1}{g_{m6} R_1 R_2 C_M}$$

In a_2 , the dominant term is

$$R_1 R_2 C_M (C_2 + C_L).$$

Thus,

$$a_2 \approx R_1 R_2 C_M (C_2 + C_L).$$

Therefore,

$$p_2 \approx -\frac{a_1}{a_2}.$$

$$p_2 \approx -\frac{C_M g_{m6} R_1 R_2}{R_1 R_2 C_M (C_2 + C_L)}.$$

$$p_2 \approx -\frac{g_{m6}}{C_2 + C_L}$$

Finally,

$$a_3 = R_C R_1 R_2 C_M C_1 (C_2 + C_L).$$

Therefore,

$$p_3 \approx -\frac{a_2}{a_3}.$$

Using

$$a_2 \approx R_1 R_2 C_M (C_2 + C_L),$$

we obtain

$$p_3 \approx -\frac{R_1 R_2 C_M (C_2 + C_L)}{R_C R_1 R_2 C_M C_1 (C_2 + C_L)}.$$

$$\boxed{p_3 \approx -\frac{1}{R_C C_1}}$$

4 T3: Compensation and Design

4.1 a

For pole-zero compensation, the zero is placed on the second pole p_2 . The zero is

$$s_z = \frac{g_{m6}}{C_M(1 - g_{m6}R_C)}.$$

$$p_2 \approx -\frac{g_{m6}}{C_2 + C_L},$$

$$\frac{g_{m6}}{C_M(1 - g_{m6}R_C)} = -\frac{g_{m6}}{C_2 + C_L}.$$

$$\boxed{R_C = \frac{1}{g_{m6}} \left(1 + \frac{C_2 + C_L}{C_M} \right)}$$